

Quantum Error Correction

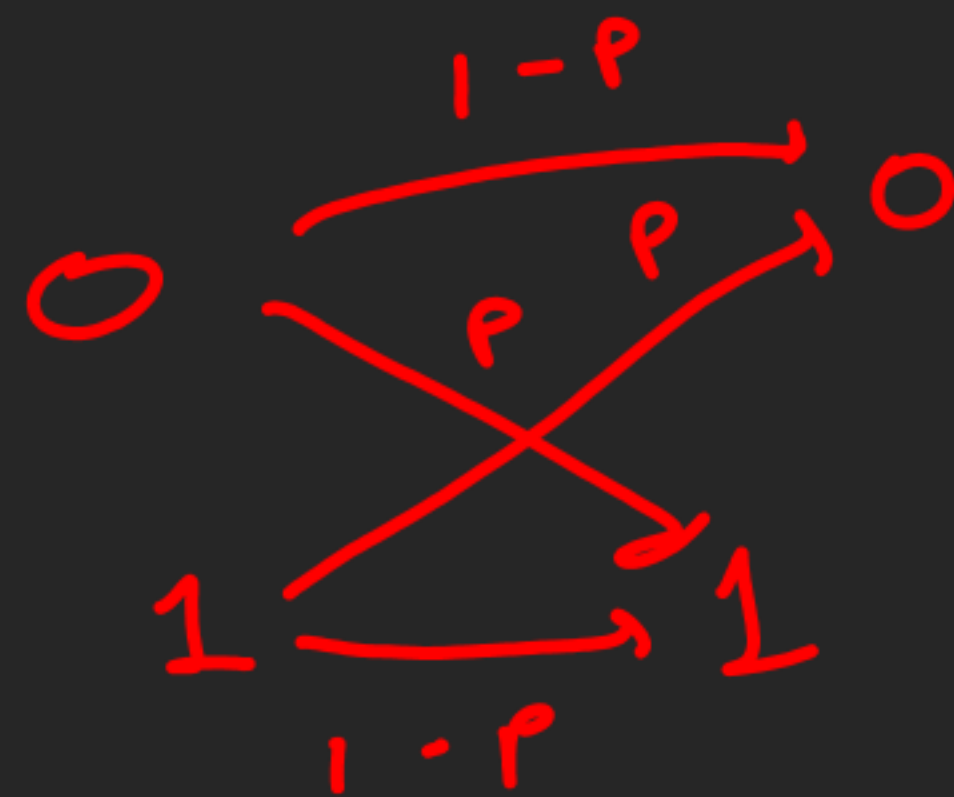
Repetition Codes

Classical Encoding and Decoding

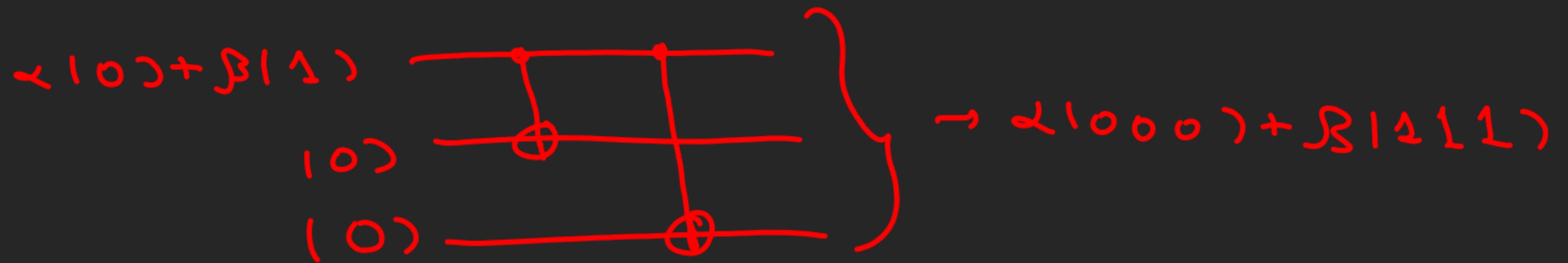
0 $\rightarrow$ 0 0 0

1 $\rightarrow$ 1 1 1

abc $\rightarrow$ majority $\{a, b, c\}$

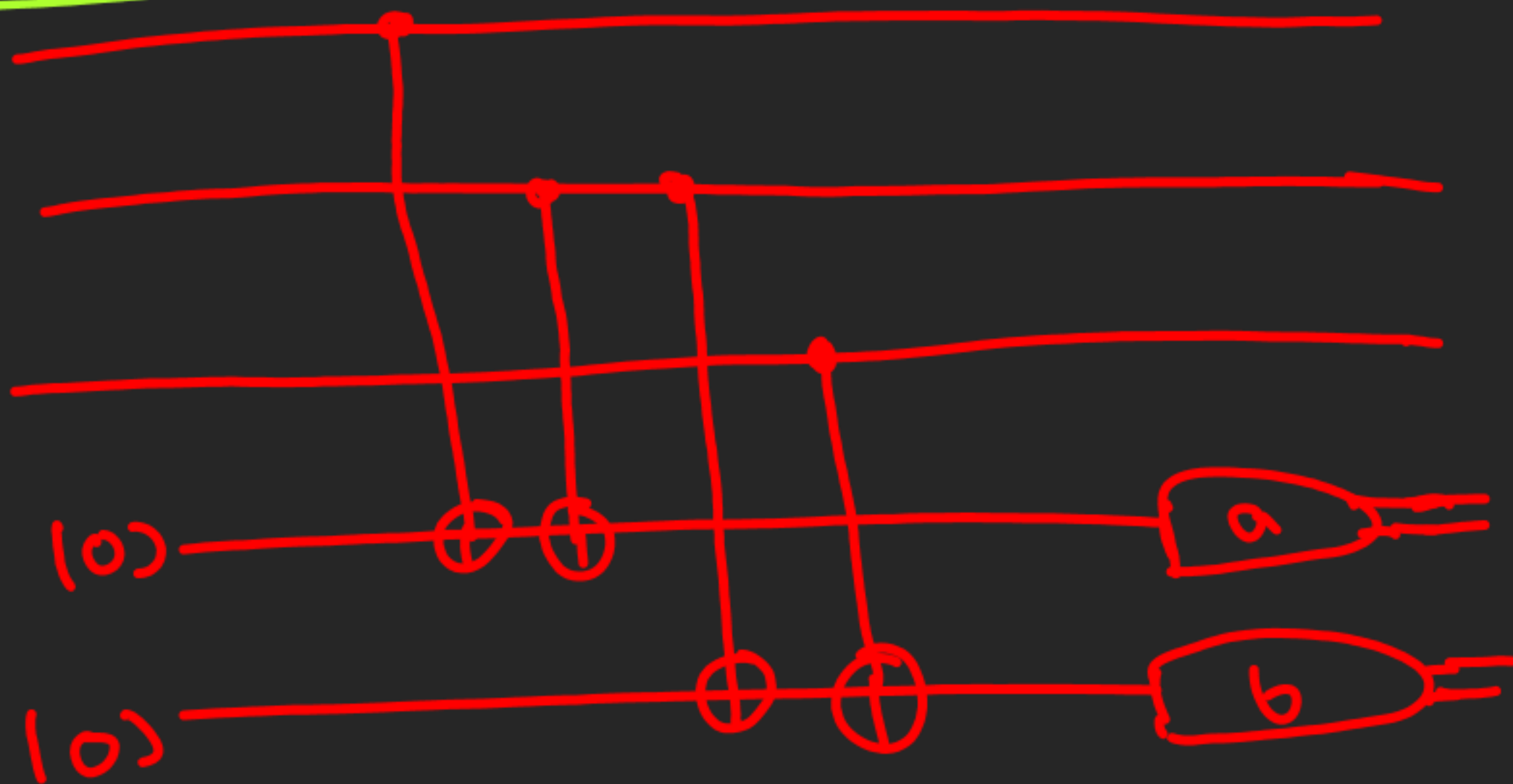


Encoding Qubits

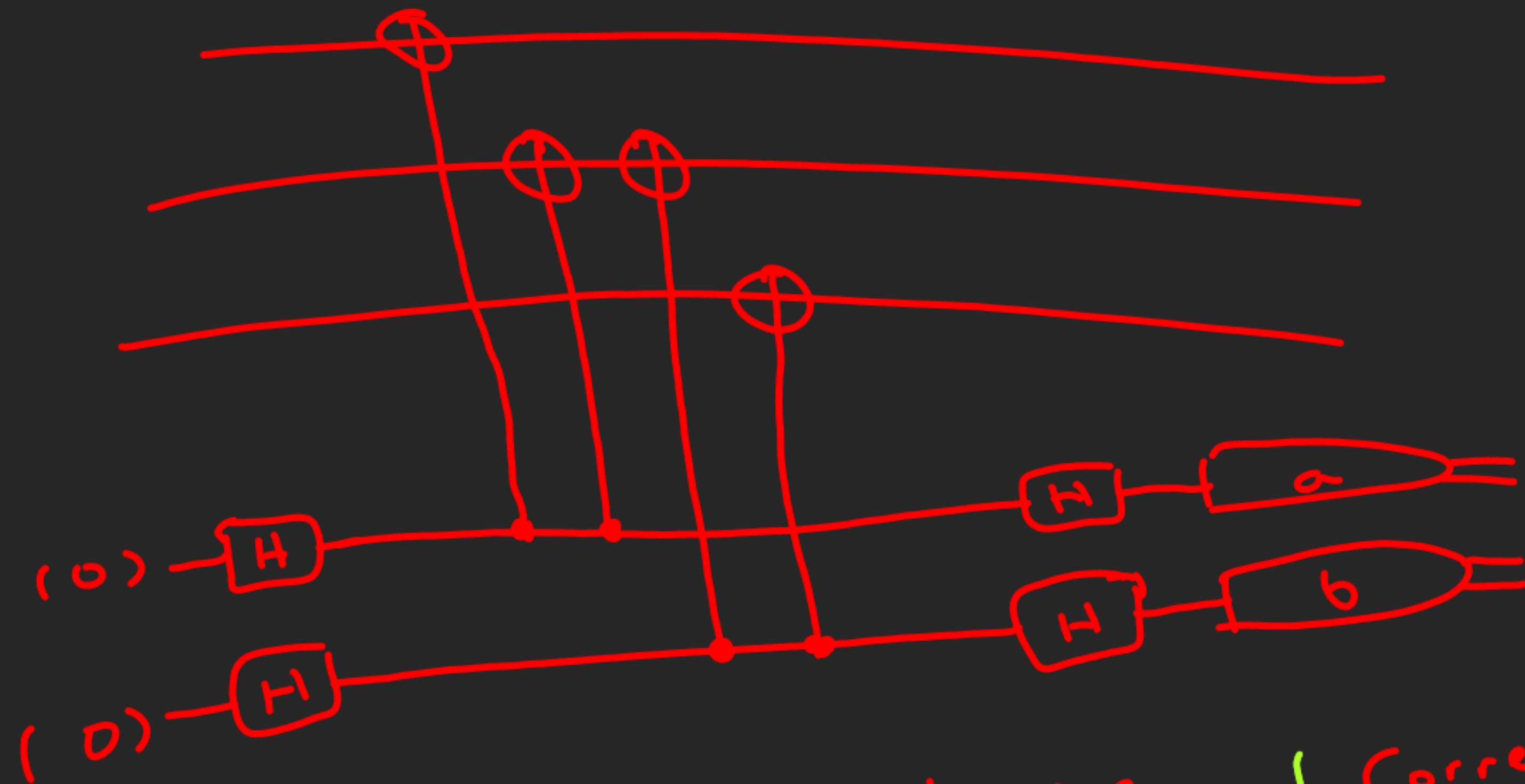


$$\alpha|0\rangle + \beta|1\rangle \rightarrow \alpha|000\rangle + \beta|111\rangle$$

Bit-Flip Error



State	Syndrome (ab)	Correction
$\alpha 1000\rangle + \beta 1111\rangle$	00	$I \otimes I \otimes I$
$\alpha 1001\rangle + \beta 1110\rangle$	01	$I \otimes I \otimes X$
$\alpha 1100\rangle + \beta 1011\rangle$	10	$X \otimes I \otimes I$
$\alpha 1010\rangle + \beta 1101\rangle$	11	$I \otimes X \otimes I$



State	Syndrome	Correction